Unpaired Echocardiogram Enhancement with CycleGAN: A Solution to the MICCAI 2025 Dehazing Challenge

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Abstract. Echocardiography is a cornerstone of cardiac imaging, yet its diagnostic utility is often compromised by noise and artifacts, particularly in patients who are difficult to image. Such degradations not only hinder clinical interpretation but also undermine the performance of AI-driven analysis pipelines. To address this challenge, we present a Cycle-Consistent Adversarial Network (CycleGAN)-based framework for enhancing and dehazing transthoracic echocardiograms. Unlike conventional supervised approaches, our method learns to map low-quality images from difficult-to-image patients to the distribution of high-quality images from easy-to-image patients without requiring paired training data. We trained the model on 4,500 four-chamber view images from 750 easy-to-image patients and 2,400 images from 40 difficult-to-image patients. By carefully balancing cycle-consistency and identity loss terms, the model effectively suppresses noise while preserving critical anatomical structures. Experimental results show substantial improvements in image quality, achieving a final score of 74.82 and a Fréchet Inception Distance (FID) of 50.91 on the test set. These findings demonstrate the promise of unpaired image-to-image translation in improving echocardiogram quality, thereby enhancing diagnostic confidence and strengthening the robustness of automated cardiac function analysis.

Keywords: CycleGAN · Image Dehazing · Echocardiography

1 Introduction

Transthoracic echocardiography (TTE) has become a cornerstone of cardiac imaging owing to its real-time acquisition, non-invasiveness, and cost-effectiveness. It allows cardiologists to evaluate essential cardiac functions and detect abnormalities that serve as critical indicators of cardiovascular disease. However, echocardiographic image quality is frequently compromised by noise and artifacts, including acoustic shadowing and signal dropout, which reduce both diagnostic reliability and interpretability. This issue is particularly severe in difficult-to-image patients, where suboptimal acoustic windows can result in images of limited clinical utility.

The need to enhance echocardiographic quality is therefore pressing. Advanced computational approaches offer the potential to improve image clarity, thereby strengthening diagnostic accuracy and enabling more robust AI-assisted analysis. In line with this objective, the Dehazing Echocardiogram Challenge at MICCAI 2025 specifically targets the enhancement and restoration of echocardiograms acquired from challenging patient populations.

The concept of generative modeling was revolutionized by Generative Adversarial Networks (GANs) [4], which introduced a novel adversarial training paradigm. This framework was notably extended to structured prediction problems like image-to-image translation for tasks where paired training data is available [5]. However, for many applications including medical imaging, obtaining such paired data is often infeasible.

In this work, we tackle this challenge by framing it as an *unpaired* image-to-image translation problem. This approach has been successfully applied to standardize ultrasound images from different domains [2]. Consequently, we leverage Cycle-Consistent Adversarial Networks (CycleGAN) [1], a powerful framework that has been specifically adapted and proven effective for ultrasound image transformation [3]. Our goal is to learn a mapping from the domain of "difficult-to-image" (hazy) echocardiograms to the domain of "easy-to-image" (clear) echocardiograms. By training on two distinct, unpaired collections of images, our model learns the underlying characteristics of high-quality images and applies these transformations to enhance the low-quality ones. We demonstrate that with specific hyperparameter adjustments, CycleGAN can effectively reduce noise and artifacts while preserving anatomical integrity, offering a promising solution for clinical application.

Our main contributions are:

- We successfully apply the CycleGAN framework to the novel task of echocardiogram dehazing for difficult-to-image patients.
- We fine-tune the CycleGAN loss functions, specifically the cycle-consistency and identity loss weights, to better suit the preservation of structural contrast in medical images.
- We validate our approach on the MICCAI 2025 challenge dataset, demonstrating its effectiveness in improving image quality.

2 Method

Our method is based on the Cycle-Consistent Adversarial Network (CycleGAN) [1], designed for image-to-image translation without paired examples. We define our source domain X as the set of echocardiograms from difficult-to-image patients and our target domain Y as the set from easy-to-image patients. Our goal is to learn a generator mapping $G : X \to Y$ that can transform any hazy image $x \in X$ into a clear image $G(x)$ that looks like it was drawn from domain Y. The network architecture follows the original paper, utilizing a generator with

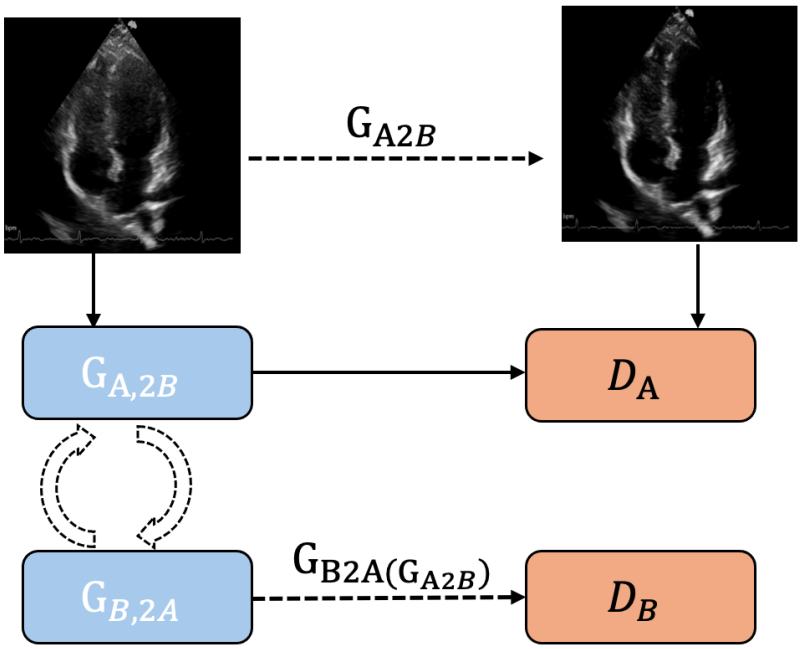

Fig. 1: CycleGAN network architecture for echocardiogram dehazing. The diagram illustrates the mapping between the hazy image domain A (from difficult-to-image patients) and the clear image domain B (from easy-to-image patients). The model uses two generators ($G_{A\to B}$, $G_{B\to A}$) and two discriminators (D_A, D_B), enforcing cycle-consistency for unsupervised learning.

residual blocks and a 70x70 PatchGAN discriminator. The overall training objective is built upon the original adversarial loss [4], augmented with techniques to ensure training stability and meaningful translations [6].

2.1 Adversarial Loss

To make the generated images indistinguishable from real images in the target domain, we employ an adversarial loss based on the original GAN formulation [4]. The generator G tries to produce realistic images $G(x)$, while a discriminator D_Y tries to differentiate between generated images $G(x)$ and real images $y \in Y$. The objective for this mapping is expressed as:

$$\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) = \mathbb{E}_{y\sim p_{\text{data}}(y)}[\log D_Y(y)] + \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log(1 - D_Y(G(x)))]$$

An analogous adversarial loss is applied for the reverse mapping $F : Y \to X$ and its corresponding discriminator D_X.

2.2 Cycle-Consistency Loss

The adversarial loss alone is insufficient to guarantee that the learned function maps an input x to a meaningful output y. To regularize the model, we introduce a cycle-consistency loss. This loss enforces the intuition that if we translate an image from domain X to Y and back to X, we should recover the original image. This forward and backward cycle consistency is captured by the following loss function:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)}[\|G(F(y)) - y\|_1]$$

This loss ensures that the generators G and F are inverses of each other, preserving the content of the image while changing its style (i.e., image quality).

2.3 Identity Loss

For tasks where we want to preserve certain characteristics of the input image, such as color or contrast, an identity loss can be beneficial. It encourages the generator to act as an identity mapping when given an image from its own target domain. For our dehazing task, this helps preserve the structural contrast of the original echo. The identity loss is defined as:

$$\mathcal{L}_{\text{identity}}(G, F) = \mathbb{E}_{y \sim p_{\text{data}}(y)}[\|G(y) - y\|_1] + \mathbb{E}_{x \sim p_{\text{data}}(x)}[\|F(x) - x\|_1]$$

2.4 Full Objective

Our full objective function combines the adversarial, cycle-consistency, and identity losses:

$$\begin{aligned}\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) + \mathcal{L}_{\text{GAN}}(F, D_X, Y, X) \\ + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}}(G, F) + \lambda_{\text{identity}}\mathcal{L}_{\text{identity}}(G, F)\end{aligned}$$

where λ_{cyc} and $\lambda_{\text{identity}}$ are hyperparameters that control the relative importance of each loss term.

3 Experiments

3.1 Dataset

We utilized the dataset provided for the MICCAI 2025 Dehazing Echocardiogram Challenge. The dataset consists of four-chamber view echocardiogram images from two distinct populations:

- **Easy-to-Image (Domain Y)**: 4500 images from 750 patients, characterized by high image quality, clear boundaries, and low noise.
- **Difficult-to-Image (Domain X)**: 2400 images from 40 patients, characterized by significant noise, artifacts, and poor contrast, which blurred cardiac structures.

The entire dataset was split into a training set and a testing set using a 9:1 ratio, ensuring that images from the same patient were kept within the same set.

3.2 Implementation Details

Our model was trained on a single NVIDIA A100 GPU for approximately 10 hours. All images were resized to 256x256 pixels and normalized to the range [-1, 1] before being fed to the networks. We used the Adam optimizer with a learning rate of 0.0002. The training was conducted for 200 epochs, with the learning rate kept constant for the first 100 epochs and linearly decayed to zero over the remaining 100 epochs.

Based on empirical observations, we adjusted the weights of the loss functions to optimize for this specific task. To encourage the model to perform a more significant transformation rather than minor adjustments, we set the cycle-consistency weight λ_{cyc} to 12 for both directions. To maintain structural integrity and contrast from the original image, we set the identity loss weight $\lambda_{\text{identity}}$ to 0.4.

3.3 Evaluation Metrics

To quantitatively evaluate the performance of our dehazing model, we used the official metrics specified by the challenge organizers:

- **Fréchet Inception Distance (FID)**: Measures the similarity of feature distributions between real and generated images. Lower is better.
- **Contrast-to-Noise Ratio (CNR & gCNR)**: Measure the image contrast. Higher is better.
- **Kolmogorov-Smirnov test (KS)**: Measures the distance between the pixel intensity distributions of hazy and dehazed images.
- **Dice & Average surface distance (ASD)**: Metrics typically used for segmentation, possibly repurposed to evaluate structural preservation.

A final composite score is calculated from these individual metrics to rank the submissions.

4 Results and Discussion

4.1 Quantitative Results

We evaluated our trained model on the held-out test set using the official evaluation script. The quantitative results are summarized in Table 1. Our method achieved a final composite score of 74.82. The FID score of 50.91 indicates a high degree of perceptual similarity between our generated images and the real clear images. Metrics such as CNR (1.43) and gCNR (0.71) confirm a significant improvement in image contrast.

Table 1: Quantitative evaluation of our method on the test set.

Metric	**Value**
Final Score	74.82
FID	50.91
CNR (mean)	1.43
gCNR (mean)	0.71
Dice (mean)	0.90
ASD (mean)	1.98
KS ROI1 (mean)	0.14
KS ROI2 (mean)	0.55

4.2 Qualitative Results

Figure 2 shows a qualitative comparison between an original hazy image and our dehazed result. As can be seen, our model effectively removes noise and enhances the visibility of key cardiac structures, such as the ventricular walls and valves. The output image exhibits sharper edges and better-defined tissue boundaries, which is crucial for accurate clinical assessment.

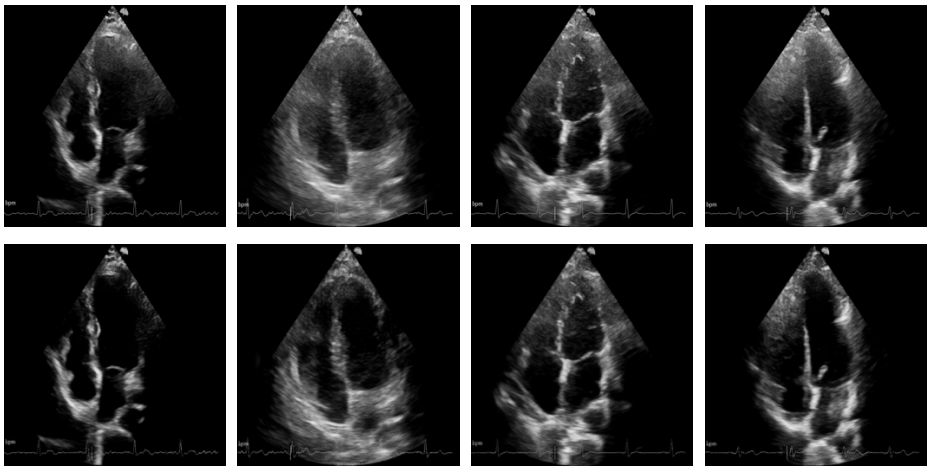

Fig. 2: Qualitative results comparing multiple examples. **Top row:** Original hazy inputs from difficult-to-image patients. **Bottom row:** Corresponding dehazed outputs from our model. Each column represents a different case, demonstrating consistent noise removal and structural enhancement across various examples.

4.3 Discussion

Our results demonstrate that the CycleGAN-based approach is highly effective for echocardiogram dehazing. The careful tuning of lambda values proved crucial. Setting the cycle-consistency weight λ_{cyc} to 12 encouraged a more substantial transformation, while the identity weight $\lambda_{\text{identity}}$ of 0.4 helped prevent the model from distorting key anatomical features. The comprehensive set of evaluation metrics confirms that our method improves the images not only in terms of perceptual quality (FID) but also in structural and contrast characteristics (CNR, gCNR, Dice, ASD).

5 Conclusion

In this paper, we presented a deep learning method based on CycleGAN for the enhancement of echocardiograms in difficult-to-image patients. By training on unpaired sets of hazy and clear images, our model learns to translate low-quality images into high-quality, diagnostically useful ones. Our approach shows strong quantitative and qualitative results on the MICCAI 2025 challenge dataset and represents a significant step towards improving the reliability of echocardiography in challenging clinical scenarios. Future work could explore incorporating anatomical priors or attention mechanisms to further refine the enhancement process.

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